

Syllabus

Nutrition in Plants, Nutrition in Animals, Respiration in Organisms, Transportation in Plants and Animals, Reproduction in Plants, Acids Bases, and Salts, Physical and Chemical Changes, Water – A Precious Resource, Heat, Motion and Time, Electric Current and Its Effects, Light.

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Science

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| <ol style="list-style-type: none"> 1. A "Pulse" is traditionally measured at the wrist (Radial artery) or the neck (Carotid artery). Why can we not feel a pulse in our veins?
 (A) Because veins do not contain any blood.
 (B) Because the blood in veins has lost all its pressure after passing through the narrow capillaries.
 (C) Because veins are located too deep inside the bones.
 (D) Because the valves in the veins block the pulse. 2. Which of the following statement is/are correct?
 I. fermentation is a chemical change
 II. Mixing vinegar and baking soda is a chemical change
 III. Condensation is a chemical change. | <ol style="list-style-type: none"> IV. Galvanised iron pipes rust easily.
 (A) IV only (B) I and II only
 (C) III and IV only (D) II and III only 3. Food left uncovered in summer becomes spoiled faster. This change is chemical because:
 (A) Temperature changes
 (B) Microorganisms produce new substances
 (C) Water evaporates
 (D) Food changes shape 4. Select the correct statement(s)
 I. An ant bite is neutralised by soap
 II. Ammonium chloride is an basic salt while sodium chloride is acidic salt
 III. The tarnished surface of copper articles can be treated by rubbing lemon on them |
|---|--|

IV. If the soil becomes too acidic, then organic matter is added to the soil which releases bases on decomposition

- (A) I and III only (B) III and II only
(C) I only (D) IV only

5. Match Column A with the correct option from Column B.

(Each option in Column B may be used once, more than once, or not at all.)

Column A	Column B
1. Heat transfer in solids	A. Radiation
2. Heat transfer through empty space	B. Black surface
3. Movement of hot and cold regions in liquids	C. Clinical thermometer
4. Device used to measure body temperature	D. Convection
5. Surface that absorbs maximum heat	E. Conduction
	F. Laboratory thermometer

(A) 1-A, 2-D, 3-E, 4-F, 5-B

(B) 1-E, 2-D, 3-A, 4-F, 5-C

(C) 1-D, 2-B, 3-A, 4-C, 5-E

(D) 1-E, 2-A, 3-D, 4-C, 5-B

6. Heavy rainfall leads to floods mainly because:

- (A) Water evaporates quickly
(B) Rivers flow uphill
(C) Excess water exceeds carrying capacity of rivers
(D) Transpiration increases

7. A solution turns methyl orange yellow and phenolphthalein pink. The solution is:

- (A) Acidic (B) Neutral
(C) Basic (D) Salty

8. A student notices that a slice of apple turns brown after being left in the open air. However, if the apple is dipped in lemon juice (Vitamin C/Acid), it stays white. What does this tell us about "Respiration" or oxidation?

(A) The lemon juice adds more sugar to the apple.

(B) The browning is an oxidation reaction (like slow respiration) caused by enzymes and oxygen; the acid in lemon juice inhibits these enzymes.

(C) The apple is breathing faster because of the lemon.

(D) The lemon juice turns the oxygen into nitrogen

9. Which of the following best represents the idea "Catch water where it falls"?

- (A) Building dams on rivers
(B) Transporting water from oceans
(C) Rooftop rainwater harvesting
(D) Using tube wells

10. If a ring of bark (including the outer layers and the soft tissue just beneath) is removed from the trunk of a living tree (girdling), the tree eventually dies. Which specific transport process is primarily blocked?

(A) The upward movement of water and minerals through the Xylem.

(B) The downward movement of food (sugar) from leaves to roots through the Phloem.

(C) The exchange of gases through the lenticels in the bark.

(D) The absorption of water by the root hairs from the soil.

11. Which of the following is the most accurate example of heat transfer by Radiation occurring in our daily lives?

(A) A spoon becoming hot when left in a cup of hot soup.

(B) The entire room becoming warm when a heater is turned on in one corner.

(C) Solar water heaters on a roof getting hot from sunlight.

(D) Boiling macaroni rising and falling in a pot of water.

12. A factory discharges acidic waste into a nearby lake. Which substance should be added to reduce the harmful effect?

(A) Sodium chloride

(B) Calcium hydroxide

(C) Hydrochloric acid

(D) Distilled water

13. During a 100-meter sprint, an athlete's leg muscles begin to ache severely. This is due to the accumulation of a specific chemical substance produced during "emergency" respiration. Identify the substance and the type of respiration.

(A) Carbon dioxide; Aerobic respiration

(B) Ethanol; Anaerobic respiration

(C) Lactic acid; Anaerobic respiration

(D) Acetic acid; Aerobic respiration

14. Observe the following changes: (i) Melting of wax, (ii) Burning of wax, (iii) Boiling of water, (iv) Rusting of iron. Which pair represents changes primarily caused by heating that are reversible (Physical Changes)?

(A) (i) and (ii) (B) (ii) and (iv)

(C) (i) and (iii) (D) (iii) and (iv)

15. Consider the following steps of nutrition in a human being:

(i) Absorption

(ii) Assimilation

(iii) Egestion

(iv) Ingestion

(v) Digestion.

Identify the correct sequence in which these processes occur in the alimentary canal.

(A) (iv) → (v) → (ii) → (i) → (iii)

(B) (iv) → (v) → (i) → (ii) → (iii)

(C) (iv) → (i) → (v) → (ii) → (iii)

(D) (v) → (iv) → (i) → (ii) → (iii)

16. Insects like cockroaches do not have blood that is red in color, nor do they use their blood to transport oxygen. Instead, they have a network of tubes. If the openings on the sides of their body were blocked with wax, the insect would die. What are these openings and tubes called?

(A) Lungs and Bronchi

(B) Spiracles and Tracheae

(C) Gills and Capillaries

(D) Stomata and Xylem

17. A laboratory thermometer shows 18°C in the morning and 42°C in the afternoon. By how much did the temperature increase?

- (A) 20°C (B) 22°C
(C) 24°C (D) 60°C

18. A gardener noticed that a plant was wilting despite being watered regularly. Upon inspection, he found that the soil was tightly compacted, and the root hairs were damaged. Why are root hairs so critical for the transportation system of the plant?

- (A) They produce food for the plant through photosynthesis in the soil.
(B) They mineral nutrients dissolved increase the surface area for the absorption of water and in water.
(C) They protect the plant from being pulled out of the ground by the wind.
(D) They prevent the Xylem from losing water back into the soil.

19. An odometer of a car reads 2000 km at the start of a trip and 2400 km at the end of the trip. If the trip took 8 hours, what was the average speed of the car in m/s?

- (A) 50 m/s (B) 13.9 m/s
(C) 500 m/s (D) 25 m/s

20. A scientist measures the pH of the fluid in different parts of the human digestive tract. She finds that the pH in the stomach is highly acidic (pH 2), but the pH in the small intestine is slightly alkaline (pH 8). Why is this change in pH necessary?

- (A) The acid in the stomach is only to taste the food.

(B) Stomach enzymes (Pepsin) need acid to work, while pancreatic enzymes in the small intestine need an alkaline medium to function.

(C) The alkaline medium helps the food to stay in the intestine for a longer time.

(D) Acidic food cannot be absorbed by the blood because it would burn the veins

21. A cheetah accelerates from rest (0 m/s) to a velocity of 20 m/s in 4 seconds. What is its acceleration?

- (A) 80 m/s^2 (B) 5 m/s^2
(C) 4 m/s^2 (D) 24 m/s^2

22. Observe a sewing machine in operation. What types of motion do the needle and the wheel exhibit respectively?



(A) Needle: Rectilinear; Wheel: Rotatory

(B) Needle: Periodic & Oscillatory; Wheel: Rotatory

(C) Needle: Rotatory; Wheel: Rotatory

(D) Needle: Curvilinear; Wheel: Periodic

23. Assertion (A): Saprotrophs, like fungi, do not have a digestive system.

Reason (R): They digest dead and decaying matter outside their body by secreting digestive juices onto it.

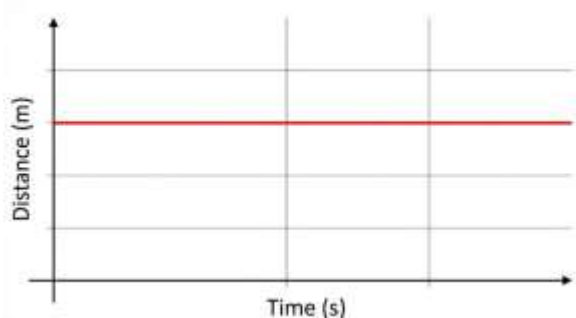
(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.

(B) Both Assertion (A) and Reason (R) are true, but R is not the correct explanation of A.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

24. Look at the Distance-Time graph of a moving object. If the graph is a straight line parallel to the Time axis (horizontal), what does this indicate about the object's motion?



(A) The object is moving with uniform speed.

(B) The object is moving with non-uniform speed.

(C) The object is at rest.

(D) The object is accelerating.

25. Equal volumes of 0.1 M HCl and 0.1 M CH_3COOH are mixed. Which statement is correct about the resulting solution?

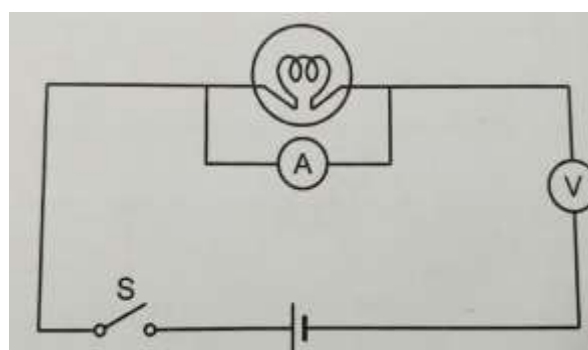
(A) It is neutral ($\text{pH} = 7$)

(B) It is acidic, with $\text{pH} < 1$

(C) It is slightly acidic, $\text{pH} > 2$

(D) It is basic

26. A student connects a circuit as shown in a diagram where an Ammeter is connected in parallel to a light bulb, and a Voltmeter is connected in series with the same bulb. What is the likely outcome when the switch is turned on?



(A) The bulb will glow very brightly.

(B) The circuit will work perfectly, giving accurate readings.

(C) The bulb will not glow because the Voltmeter has very high resistance.

(D) The voltmeter will burn out immediately.

27. A bottle of grape juice is left in a warm corner for a week. When opened, it fizzes and smells of alcohol. Which organism and which process are responsible for this change?

(A) Bacteria; Aerobic Respiration.

(B) Yeast; Anaerobic Respiration (Fermentation).

(C) Algae; Photosynthesis.

(D) Virus; Reproduction

28. Assertion (A): Exhaled air contains a higher percentage of Carbon Dioxide than inhaled air.

Reason (R): Carbon Dioxide is produced as a byproduct when food is broken down in the presence of oxygen in cells.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.

(B) Both Assertion (A) and Reason (R) are true, but R is not the correct explanation of A.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

29. Calculate the electrical energy consumed (in kWh) if a 1000 Watt electric iron is used for 2 hours.

- (A) 2000 kWh (B) 2 kWh
(C) 500 kWh (D) 0.5 kWh

30. An electric bell works on the principle of electromagnetism. Which component acts as the "Make and Break" mechanism to keep the hammer striking repeatedly?

- (A) The Gong
(B) The Electromagnet coil
(C) The Interrupter (Contact Screw and armature strip)
(D) The Battery

31. In a diagram of the human respiratory system, what is the specific function of the 'Alveoli'?



(A) To produce mucus to trap dust.

(B) To act as a site for the exchange of gases between air and blood.

(C) To vibrate and produce sound.

(D) To prevent the trachea from collapsing.

32. Fluoride contamination in water can lead to:

- (A) Goiter
(B) Dental and skeletal fluorosis
(C) Hemophilia
(D) Night blindness

33. If 10 Coulombs of charge flows through a wire in 5 seconds, what is the current flowing through the wire?

- (A) 50 Amperes (B) 2 Amperes
(C) 0.5 Amperes (D) 10 Amperes

34. During a "Cold Shower," a person often takes a sudden, sharp gasp of air. This is an involuntary reflex. What is the body trying to achieve?

- (A) To warm up the water using the heat from the lungs.
(B) To increase the oxygen supply as the heart rate increases due to the "cold shock" response.
(C) To prevent water from entering the ears.
(D) To stop the digestion process temporarily.

35. A narrow beam of light strikes a plane mirror. The angle between the incident ray and the surface of the mirror (the glancing angle) is 35° . What is the angle of reflection?

- (A) 35° (B) 55°
(C) 70° (D) 90°

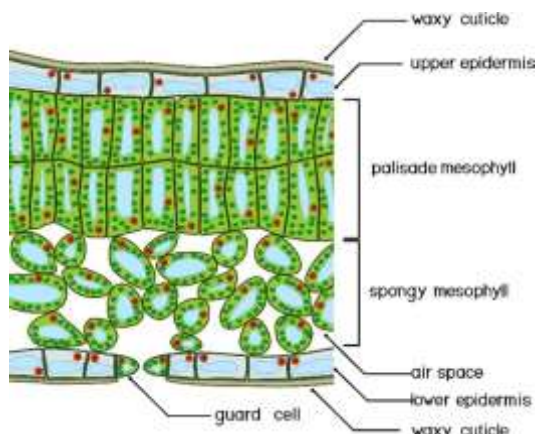
36. Srujana tested the nature of three different substances with different indicators and recorded her observations as shown:

- Substances l turned Methyl Orange indicator red
- Substances m turned China rose indicator green.
- Substances n turned turmeric indicator red

The substances l, m, and n are ____.

- (A) Sugar Solution, Spinach juice, Baking soda
(B) Window Cleaner, Apple juice Soap Solution
(C) Lemon Juice, Window Cleaner, Milk of magnesia
(D) Lime water, Milk of magnesia, Curd

37. A diagram shows a cross-section of a leaf. The 'X' represents pores located mostly on the underside. If these pores are blocked with grease, which process will be most affected?



- (A) Transpiration and Gas Exchange
(B) Upward movement of food
(C) Absorption of minerals from soil
(D) Germination of seeds

38. Consider the following light sources:

- (i) The Sun,
(ii) A lit electric bulb,
(iii) A laser pointer,
(iv) A glowing firefly. Which classification regarding these sources is logically correct?

(A) (i) and (iii) are Natural Luminous sources; (ii) is Artificial Luminous.

(B) (i) and (iv) are Natural Luminous sources; (ii) and (iii) are Artificial Luminous.

(C) Only (i) is a Natural Luminous source; the rest are Artificial.

(D) All are Non-Luminous objects reflecting light.

39. A person consumes a "Sugar-Free" candy that contains a complex carbohydrate the human body cannot digest. What will be the most likely path of this carbohydrate in the digestive system?

(A) It will be absorbed into the blood in the stomach.

(B) It will be broken down by salivary amylase and used for energy.

(C) It will pass through the small intestine undigested and be fermented by bacteria in the large intestine.

(D) It will be converted into proteins by the liver.

40. An object is placed 2 meters away from a plane mirror. If the object is moved 50

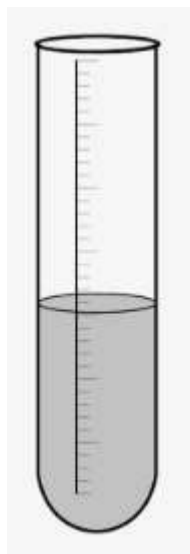
cm towards the mirror, what will be the new distance between the object and its image?

- (A) 1.5 meters (B) 2.5 meters
(C) 3.0 meters (D) 4.0 meters

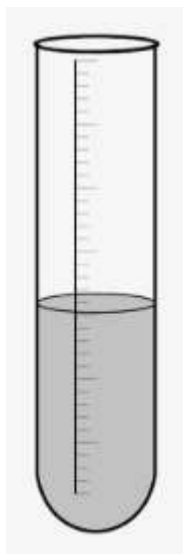
41. A person is suffering from a condition called "Steatorrhea," where their stools contain large amounts of undigested fat. This could be due to a blockage in the duct of which of the following organs?

- (A) Salivary glands
(B) Pancreas and Gallbladder
(C) Large Intestine
(D) Esophagus

42. Select the correct Statements regarding solutions given in test tubes 1 and 2



(1) Lemon
Solution



(2) Detergent
Solution

I. Solution 1 will turn methyl orange indicator red

II. Solution 2 will turn turmeric indicator red

III. Solution 1 will turn methyl orange indicator yellow

IV. Solution 2 will turn China rose indicator magenta

- (A) I and II only (B) II and III only
(C) II only (D) IV only

43. A farmer notices reduced rainfall in his area after deforestation. Which process is most affected?

- (A) Condensation (B) Evaporation
(C) Transpiration (D) Precipitation

44. Which of the following reactions can be reversed

- (A) steam condensing back into liquid water.
(B) Raw egg to boiled egg
(C) Photo synthesis
(D) Burned paper

45. Read the given statements and select the correct option

Statement 1: CO_2 gas when dissolved in water turns Phenolphthalein indicator into colourless

Statement II: CO_2 is a non – metallic oxide and forms acidic solution in water

(A) Both statement 1 and Statement 2 are true and Statement 2 is the correct explanation of Statement 1

(B) Both statement 1 and Statement 2 are true and Statement 2 is not the correct explanation of Statement 1

(C) Statement 1 is true but Statement 2 is false

(D) Both Statement 1 and Statement 2 are false

46. Why does distilled water not conduct electricity but rainwater does?

- (A) Rainwater contains dissolved salts and ions
 - (B) Distilled water contains dissolved salts
 - (C) Distilled water contains more impurities than rainwater
 - (D) Rainwater does not contain any ions
- 47.** Why are big cities often located near rivers?
- (A) Rivers reduce evaporation
 - (B) Rivers provide abundant freshwater
 - (C) Rivers prevent droughts
 - (D) Rivers increase rainfall
- 48.** Imagine you have a sheet of paper with a small pinhole in the center. You hold it between a lit candle and a white screen. What do you observe on the screen?
- (A) An upright, blurred image of the flame.
 - (B) An inverted, sharp image of the flame.
 - (C) A shadow of the flame.
 - (D) No image is formed.
- 49.** An experimenter puts a slice of bread in a sealed plastic bag with a small packet of "Oxygen Absorber." A week later, no mold has grown. In a second bag

without the absorber, mold is everywhere.

What does this apply about the requirements for fungal nutrition?

(A) Fungi need oxygen for aerobic respiration to break down the food they absorb from the bread.

(B) Fungi only grow in the dark.

(C) The oxygen absorber acts as a poison to the fungi.

(D) Bread cannot be digested if oxygen is present.

50. Which of the following is not a step in water treatment for drinking?

- (A) Coagulation (B) Sedimentation
- (C) Chlorination (D) Electrolysis

KEY

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|-------|-------|-------|-------|-------|-------|-------|
| 1) B | 2) B | 3) B | 4) A | 5) D | 6) C | 7) C |
| 8) B | 9) C | 10) B | 11) C | 12) B | 13) C | 14) C |
| 15) B | 16) B | 17) C | 18) A | 19) B | 20) B | 21) B |
| 22) B | 23) A | 24) C | 25) C | 26) C | 27) B | 28) A |
| 29) B | 30) C | 31) B | 32) B | 33) B | 34) B | 35) B |
| 36) C | 37) A | 38) B | 39) C | 40) C | 41) B | 42) A |
| 43) C | 44) A | 45) A | 46) A | 47) B | 48) B | 49) A |
| 50) D | | | | | | |